

Assignment-2 Advance Linear Algebra

1. Let $V = \{(x, y) : x, y \in \mathbb{R}\}$. Show that $V_{\mathbb{R}}$ is not a vector space under the following operations:

- (a) $(x, y) + (z, t) = (x, y)$ and $a(x, y) = (ax, ay)$
- (b) $(x, y) + (z, t) = (x + z, y + t)$ and $a(x, y) = (ax, y)$
- (c) $(x, y) + (z, t) = (x + z, 0)$ and $a(x, y) = (ax, ay)$
- (d) $(x, y) + (z, t) = (x + z, y + t)$ and $a(x, y) = (a^2x, a^2y)$

2. Let V be the set of positive real numbers. We define vector addition and scalar multiplication as:

$$x + y = xy \quad \text{and} \quad ax = x^a$$

for all $x, y \in V$ and $a \in \mathbb{R}$. Prove that V is a vector space.

3. Let $S = \{(1, 1, 0), (0, 2, 0)\}$. Show that

$$W = \{(a, b, 0) \mid a, b \in \mathbb{R}\}$$

is the subspace of $\mathbb{R}^3(\mathbb{R})$ generated by S .

4. If $S = \{(1, 2, 3, 4), (1, -1, 1, 0), (1, 2, 1, -1)\}$ and $T = \{(3, 3, 5, 3), (1, -1, 3, 5), (3, 0, 5, 4), (2, 1, 4, 4)\}$ are subsets of the vector space $\mathbb{R}^4$, determine whether or not $L(S)$ and $L(T)$ are equal.

5. Verify the Rank-Nullity theorem for the linear transformation

$$T : \mathbb{R}^4 \rightarrow \mathbb{R}^3 \quad \text{defined by} \quad T(x, y, z, w) = (x + y, y - z, z - w).$$

6. Use the Gram-Schmidt process to find an orthogonal basis for the vector space $\mathbb{R}^3$ with the standard inner product, starting from the given basis:

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

7. Let W be the subset of $\mathbb{R}^3$ defined by:

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \mid 2x - y + z = 0 \right\}.$$

Prove that W is a subspace of $\mathbb{R}^3$ and find its dimension.

8. Find all the eigenvalues and a basis for each eigenspace of the linear operator $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by:

$$T(a, b, c) = (3a + b + 4c, 2b + 6c, 5c).$$

Is it diagonalizable? If so, find a basis B for $\mathbb{R}^3$ such that $[T]_B$ is a diagonal matrix.

9. Prove that

$$B = \{(1, 1, 0, 0), (0, 1, 1, 0), (0, 0, 1, 1), (1, 0, 0, 0)\}$$

is a basis of $\mathbb{R}^4(\mathbb{R})$ and determine the coordinate of $(2, 3, 4, -1)$ relative to the ordered basis B .

10. Show that $\mathbb{Q}$, the set of rational numbers, is a vector space over $\mathbb{R}$.